- Deploy the Dapp on the blockchain client node in a cloud-environment such as Infura.
- Design a simple, intuitive UI for development purposes. This design will enable and guide the team later to develop a production-quality UI (a topic that is beyond the scope of this book).
- Test the integrated Dapp thoroughly before distributing it to stakeholders. Distribute only the client application part to the peer participants, who may be using a lightweight web client or mobile client.

Summary

- The blockchain-based solution is typically part of a larger system. In this chapter, DCC-Dapp for facilitating a certificate program is part of a larger university system.
- A roadmap is helpful for analyzing, designing, and developing a blockchain-based decentralized application solution for a problem.
- Analyzing a problem for its roles, rules, data structures, functions, and events emitted guides the development of a smart contract.
- A single complex off-chain operation for adding courses is split into four smaller smart contract functions (add prerequisite, add core, and so on) for simpler transactions to be recorded on-chain.
- JavaScript testing scripts and the Truffle test tool help in automating smart contract testing.
- Dapp development begins with smart contract development, its local deployment, and testing; migrating it to a production infrastructure; and creating a deployable module for the participants.